

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2004

MATHEMATICS EXTENSION 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks - 84

- Attempt Questions 1 – 7
- All questions are of equal value

Name: _____

Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Total

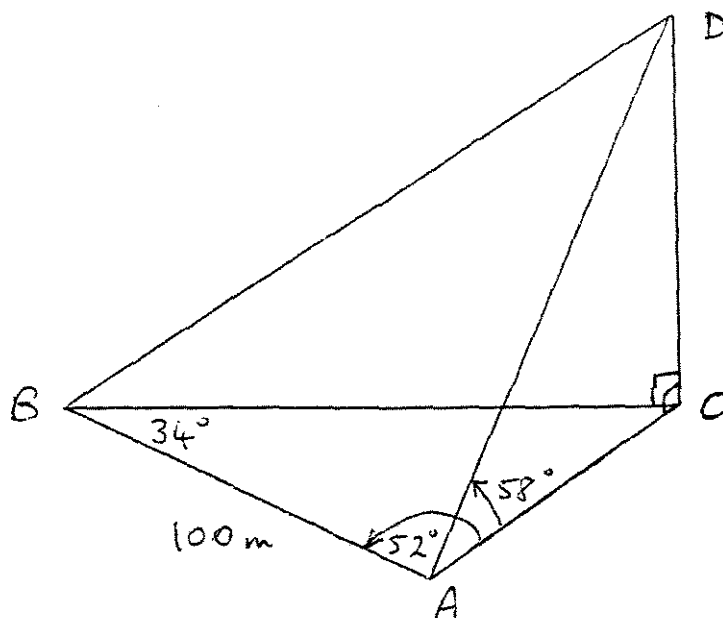
Question 1

- a) Simplify $\frac{1+a^{-1}}{1+a^{-3}}$ 2
- b) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$ 2
- c) Find $\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{2x^2}$ 2
- d) Use the substitution $u = 1 + x^3$, or otherwise to evaluate $\int_0^1 x^2(1+x^3)^3 dx$ 4
- e) Find the acute angle between the lines $x + y\sqrt{3} = 3$ and $y = 3$ 2

Question 2 (Start a new page)

- a) One of the roots of $2x^3 + x^2 - 15x - 18 = 0$ is positive and equal to the product of the other two roots. Find this root. 2
- b) If $\frac{dy}{dx} = 1 + y$, and when $x = 0, y = 2$; show that $y = 3e^x - 1$ 3
(hint: examine $\frac{dx}{dy}$.)
- c) Find $\int \frac{dx}{\sqrt{16 - 25x^2}}$ 3

d)



A pole DC is seen from two points A and B. The angle of elevation from A is 58° , $\angle CAB$ is 52° , $\angle ABC$ is 34° and A and B are 100m apart. Find:

- | | | |
|------|--|---|
| (i) | How far A is from the foot of the pole, to the nearest metre | 3 |
| (ii) | The height of the pole, to the nearest metre | 1 |

Question 3 (Start a new page)

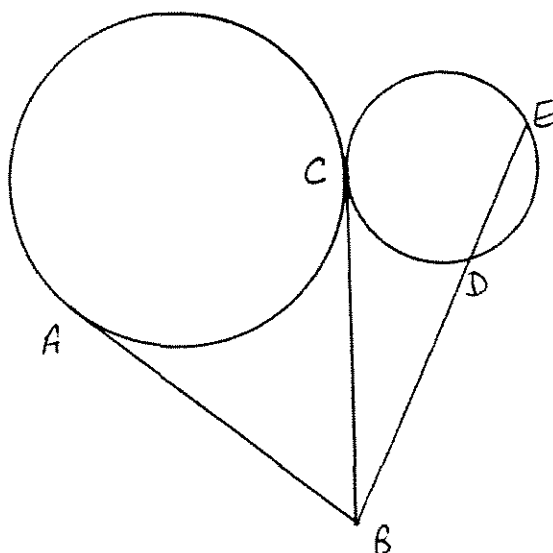
- | | | |
|-------|---|---|
| a) | The equation $\sin \theta + \theta - 2 = 0$ has a root near $\theta = 1.1$. Use this as a first approximation and one application of Newton's Method to find a better approximation of the root correct to 3 decimal places. | 3 |
| b) | $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. | |
| (i) | Find the coordinates of M, the midpoint of PQ | 1 |
| (ii) | If the gradient of PQ is constant, find the equation for the locus of M and show that it is a line parallel to the axis of the parabola. | 3 |
| c) | Given the function $f(x) = 1 - \tan x$ for the domain $0 \leq x \leq \frac{\pi}{4}$: | |
| (i) | Sketch the graph of $y=f(x)$ | 1 |
| (ii) | Show that $\int \tan x \, dx = -\ln(\cos x) + c$ | 1 |
| (iii) | The region in (i) is rotated about the x axis. Find the volume of the solid generated to 2 decimal places. | 3 |

Question 4 (Start a new page)

- a) Find $\int \cos^2 2x \, dx$ 2
- b) Prove by Mathematical Induction, that for all positive integers n : 4
- $$\sum_{r=1}^n r(r+1) = \frac{n(n+1)(n+2)}{3}$$
- c) The displacement x cm of an object from the origin is given by
- $$x = \cos t - \sqrt{3} \sin t$$
- (i) Prove that the object executes simple harmonic motion. 2
- (ii) Find an exact time when the object reaches maximum speed 1
- (iii) Express the displacement in the form $A \cos(nt + \alpha)$ and state the amplitude. 3

Question 5 (Start a new page)

a)



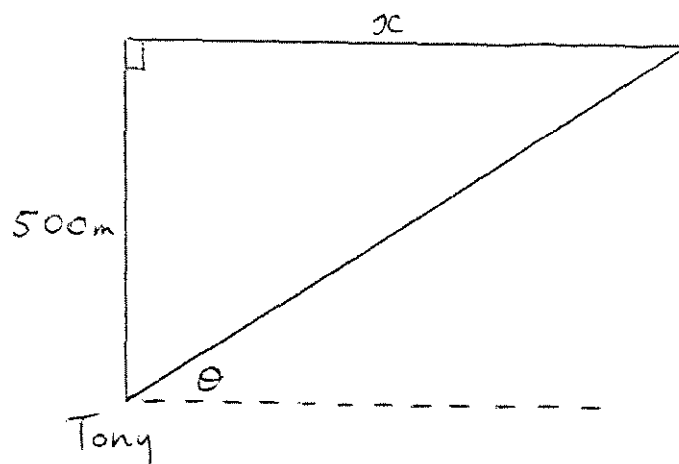
Not to scale

AB and BC are tangents
and $BD = 4 DE$
Prove that $AB = 2\sqrt{5} DE$,
giving reasons.

2

- b) The acceleration of a body P is given by $\frac{d^2x}{dt^2} = 18x(x^2 + 1)$, where x is the displacement of P from 0 at time t . The velocity is v .
Given $t = 0, x = 0, v = 3$ and that $v > 0$ throughout the motion:
- (i) find v in terms of x 2
- (ii) show that $x = \tan 3t$ 2

c)



At 9am, an ultralight aircraft flies directly over Tony's head at a height of 500m. It maintains a constant speed of 20 m/s and a constant altitude.

If x is the horizontal distance travelled by the plane and θ is the angle of elevation from Tony to the plane,:

- (i) Show that $\frac{dx}{d\theta} = -500 \operatorname{cosec}^2 \theta$ 2
- (ii) Hence show that $\frac{d\theta}{dt} = \frac{-1}{25} \sin^2 \theta$ 2
- (iii) Find the rate of change of the angle of elevation at 9.01am (in radians per second) 2

Question 6 (Start a new page)

- a) ABCD is a cyclic quadrilateral.
Show that $\tan A + \tan B + \tan C + \tan D = 0$ 2
- b) A sky-diver opens his parachute when falling at 30 m/s. Thereafter, his acceleration is given by $\frac{dv}{dt} = k(6 - v)$ where k is a constant.
 - (i) Show that this differential equation is satisfied by $v = 6 + Ae^{-kt}$ and find the value of A . 2
 - (ii) One second after opening his chute, his velocity is 10.7 m/s. Find the value of k to 2 decimal places. 1
 - (iii) Find his velocity, correct to one decimal place, two seconds after his chute is opened. 1

- c) A soldier is 150 metres from, and on the same horizontal level as, her target. Her weapon can fire with an initial velocity of 50 m/s. Take $g = 10 \text{ m/s}^2$.
- (i) Write the equations of motion for horizontal and vertical displacement. 2
- (ii) Find the two possible angles at which she must fire her weapon to hit the target. 4

Question 7 (Start a new page)

- a) The function $f(x) = \sec x$ is defined for $0 \leq x < \frac{\pi}{2}$.
- (i) State the domain of the inverse function $f^{-1}(x)$. 1
- (ii) Show that $f^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$ 1
- (iii) Hence find $\frac{d}{dx}[f^{-1}(x)]$ 2
- b) (i) Find all real solutions to the equation $x^4 + x^2 - 1 = 0$, giving your answers correct to three decimal places. 2
- (ii) On the same axes, sketch the graphs of $y = \tan^{-1} x$ and $y = \cos^{-1} x$. Label important points. Mark the point P where the two curves intersect. 2
- (iii) If $\tan^{-1} x = \cos^{-1} x$ at P, show that $x^4 + x^2 - 1 = 0$ and find the coordinates of P. 4

SOLUTIONS.

$$\textcircled{1} a) \frac{1 + \frac{1}{a}}{1 + \frac{1}{a^3}} = \frac{\frac{a+1}{a}}{\frac{a^3+1}{a^3}} \leftarrow \textcircled{1}$$

$$= \frac{a+1}{a} \times \frac{a^3}{a^3+1} \leftarrow \textcircled{1}$$

$$= \frac{a+1}{a} \times \frac{a^3}{(\cancel{a+1})(a^2-a+1)}$$

$$= \frac{a^2}{a^2-a+1}$$

$$b) \frac{d}{dx}[(\cos x)^{-1}] = -(\cos x)^{-2} \cdot (-\sin x) \leftarrow \textcircled{1}$$

$$= \frac{\sin x}{\cos^2 x} \leftarrow \textcircled{1}$$

$$= \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$$

$$= \sec x \tan x$$

$$c) = \lim_{x \rightarrow 0} \frac{\sin^2 x}{2x^2} \leftarrow \textcircled{1}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2$$

$$= \frac{1}{2} \times 1 \leftarrow \textcircled{1}$$

$$= \frac{1}{2}$$

$$d) \int_0^1 x^2(1+x^3)^3 dx = \int_1^2 x^2(u^3) \frac{du}{3x^2}$$

$$= \frac{1}{3} \int_1^2 u^3 du \textcircled{1}$$

$$= \frac{1}{3} \left[\frac{u^4}{4} \right]$$

$$= \frac{1}{3} \left(4 - \frac{1}{4} \right) = 1 \frac{1}{4} \textcircled{1}$$

$$\begin{aligned} u &= 1+x^3 \\ \frac{du}{dx} &= 3x^2 \\ dx &= \frac{du}{3x^2} \textcircled{1} \\ x=0, u &= 1 \textcircled{1} \\ x=1, u &= 2 \end{aligned}$$

$$e) y = -\frac{x}{\sqrt{3}} + 3, m_1 = -\frac{1}{\sqrt{3}}$$

$$m_2 = 0$$

$$\tan \theta = \left| \frac{-\frac{1}{\sqrt{3}} - 0}{1+0} \right| \textcircled{1}$$

$$= \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ \textcircled{1}$$

$$\textcircled{2} a) \text{ let roots be } \alpha, \beta, \alpha\beta$$

$$\text{and } \alpha \cdot \beta \cdot \alpha\beta = -\frac{d}{a} \textcircled{1}$$

$$\therefore \alpha^2 \beta^2 = 9$$

$$\therefore \alpha\beta = 3 (>0) \textcircled{1}$$

$$b) \frac{dx}{dy} = \frac{1}{1+y}$$

$$\therefore x = \log(1+y) + c \textcircled{1}$$

$$\text{Sub } x=0, y=2 :$$

$$\therefore 0 = \log 3 + c$$

$$\therefore c = -\log 3$$

$$\therefore x = \log(1+y) - \log 3 \textcircled{1}$$

$$= \log\left(\frac{1+y}{3}\right)$$

$$\therefore e^x = \frac{1+y}{3} \textcircled{1}$$

$$\therefore 3e^x = 1+y$$

$$\therefore y = 3e^x - 1$$

$$\begin{aligned}
 \text{c) } \int \frac{dx}{\sqrt{16-25x^2}} &= \int \frac{dx}{\sqrt{25(\frac{16}{25}-x^2)}} \quad (1) \\
 &= \frac{1}{5} \int \frac{dx}{\sqrt{(\frac{4}{5})^2 - x^2}} \quad (1) \\
 &= \frac{1}{5} \sin^{-1} \frac{5x}{4} + c \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{d) (i) } \frac{AC}{\sin 34^\circ} &= \frac{100}{\sin 94^\circ} \quad \leftarrow (1) \text{ for rule} \\
 \therefore AC &= \frac{100 \sin 34^\circ}{\sin 94^\circ} \quad (1) \\
 &= 56 \text{ m (nearest m)} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \tan 58^\circ &= \frac{DC}{56} \\
 \therefore DC &= 56 \tan 58^\circ \\
 &= 90 \text{ m (nearest m)} \quad (1)
 \end{aligned}$$

$$(3) \text{ a) } f'(\theta) = \cos \theta + 1 \quad (1)$$

$$\begin{aligned}
 \therefore a_2 &= 1.1 - \frac{f(1.1)}{f'(1.1)} \\
 &= 1.1 - \frac{\sin 1.1 + 1.1 - 2}{\cos 1.1 + 1} \quad (1) \\
 &\doteq 1.106 \quad (1)
 \end{aligned}$$

$$\text{b) (i) } \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$$

$$\therefore M \text{ is } \left[a(p+q), \frac{a}{2}(p^2+q^2) \right] \quad (1)$$

$$\text{(ii) } M_{pq} = c$$

$$\therefore \frac{aq^2 - ap^2}{2aq - 2ap} = c \quad (1)$$

$$\therefore \frac{(q-p)(q+p)}{2(q-p)} = c$$

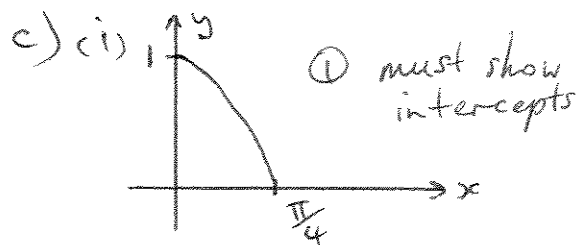
$$\therefore \frac{q+p}{2} = c \Rightarrow q+p = 2c \quad (1)$$

$\therefore M$ has coords

$$\left[2ac, \frac{a}{2}(p^2+q^2) \right]$$

$$\therefore x = 2ac \quad (1)$$

$\therefore M$ has locus eqn $x = 2ac$, which is vertical and parallel to axis of parabola.



$$\begin{aligned}
 \text{(ii) } \int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx \quad (1) \\
 &= -\ln(\cos x) + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) } V &= \pi \int_0^{\pi/4} (1 - \tan x)^2 \, dx \\
 &= \pi \int_0^{\pi/4} (1 - 2\tan x + \tan^2 x) \, dx \\
 &= \pi \int_0^{\pi/4} (\sec^2 x - 2\tan x) \, dx \quad (1) \\
 &= \pi \left[\tan x + 2 \log(\cos x) \right]_0^{\pi/4} \quad (1) \\
 &\doteq 0.96 \quad (1)
 \end{aligned}$$

④ a) $\int \frac{1}{2}(1 + \cos 4x) dx$ ①
 $= \frac{1}{2} \left(x + \frac{\sin 4x}{4} \right) + C$ ①

b) Prove true for $n=1$:

LHS = $1 \times 2 = 2$

RHS = $\frac{1 \cdot 2 \cdot 3}{3} = 2 = \text{LHS}$ ①

$\therefore$ result is true for $n=1$

Assume true for $n=k$:

ie. assume $S_k = \frac{k(k+1)(k+2)}{3}$ ①

Prove true for $n=k+1$:

ie. prove $S_{k+1} = \frac{(k+1)(k+2)(k+3)}{3}$ ①

Now, $S_{k+1} = S_k + T_{k+1}$

① $\left\{ \begin{aligned} &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= \frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3} \\ &= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\ &= \frac{(k+1)(k+2)(k+3)}{3} \end{aligned} \right.$ shown

Since the result is true for $n=1$, then, from above, it must be true for $n=1+1=2$ and $n=2+1=3$ and so on for all positive integers n

c) (i) $\frac{dx}{dt} = -\sin t - \sqrt{3} \cos t$
 $\frac{d^2x}{dt^2} = -\cos t + \sqrt{3} \sin t$
 $= -x$ ①

which is in the form

$\ddot{x} = -n^2 x$ for SHM ①

(ii) max. speed when $\ddot{x} = 0$

$\therefore -\cos t + \sqrt{3} \sin t = 0$

$\therefore \sqrt{3} \sin t = \cos t$

$\therefore \tan t = \frac{1}{\sqrt{3}} (\cos t \neq 0)$

$\therefore t = \frac{\pi}{6}$ seconds ①
 (or equiv.)

(iii) $A = \sqrt{1 + (\sqrt{3})^2}$ and $n=1$
 $= \sqrt{4}$ from (i)
 $= 2$

$\therefore \cos t - \sqrt{3} \sin t = 2 \cos(t + \alpha)$

$\therefore \frac{1}{2} \cos t - \frac{\sqrt{3}}{2} \sin t = \cos(t + \alpha)$
 $= \cos t \cos \alpha - \sin t \sin \alpha$

$\therefore \left. \begin{aligned} \cos \alpha &= \frac{1}{2} \\ \sin \alpha &= \frac{\sqrt{3}}{2} \end{aligned} \right\} \therefore \alpha = \frac{\pi}{3}$ ① ①

$\therefore \cos t - \sqrt{3} \sin t = 2 \cos(t + \frac{\pi}{3})$

$\therefore$ amplitude = 2 units. ①

⑤

a) $BC^2 = BD \cdot BE$

(square of tangent = product of intersecting chords)

$= 4DE \times 5DE$ ①
 $= 20DE^2$

$\therefore BC = \sqrt{20DE^2}$
 $= 2\sqrt{5} DE$

and $AB = BC$ (equal tangents to a circle) ①

$\therefore AB = 2\sqrt{5} DE$

$$b)(i) \frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 18x^3 + 18x$$

$$\therefore \frac{1}{2} v^2 = \frac{18x^4}{4} + 9x^2 + c$$

$$\therefore v^2 = 9x^4 + 18x^2 + k \quad (1)$$

$$(x=0, v=3):$$

$$\therefore 9 = 0 + 0 + k \quad (k=9)$$

$$\therefore v^2 = 9x^4 + 18x^2 + 9$$

$$= (3x^2 + 3)^2$$

$$\therefore v = 3x^2 + 3 (>0) \quad (1)$$

$$(ii) v = \frac{dx}{dt} = 3x^2 + 3$$

$$\therefore \frac{dt}{dx} = \frac{1}{3x^2 + 3}$$

$$\therefore t = \frac{1}{3} \int \frac{1}{x^2 + 1} dx \quad (1)$$

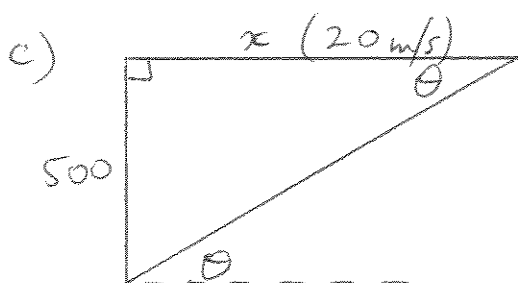
$$= \frac{1}{3} \tan^{-1} x + c$$

$$(t=0, x=0):$$

$$\therefore 0 = 0 + c \quad (c=0)$$

$$\therefore 3t = \tan^{-1} x \quad (1)$$

$$\therefore \tan 3t = x \text{ as reqd.}$$



$$(i) \tan \theta = \frac{500}{x}$$

$$\therefore x = \frac{500}{\tan \theta} \quad (1)$$

$$\therefore \frac{dx}{d\theta} = \frac{-\sec^2 \theta \times 500}{\tan^2 \theta}$$

$$= \frac{-500}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta} \quad (1)$$

$$= \frac{-500}{\sin^2 \theta}$$

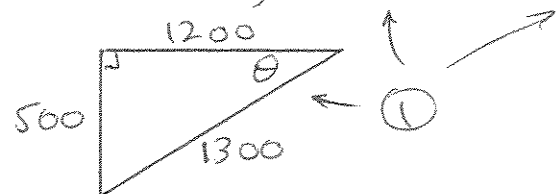
$$= -500 \operatorname{cosec}^2 \theta$$

$$(ii) \frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt} \quad (1)$$

$$= \frac{\sin^2 \theta}{-500} \times 20 \quad (1)$$

$$= -\frac{1}{25} \sin^2 \theta$$

$$(iii) \text{ At 9.01 am, } t=60, x=1200 \text{ m}$$



$$\therefore \frac{d\theta}{dt} = -\frac{1}{25} \times \left(\frac{5}{13} \right)^2$$

$$= -\frac{1}{25} \times \frac{25}{169}$$

$$= -\frac{1}{169} \text{ radians/second} \quad (1)$$

$$(0.006)$$

$$(6) a) \text{ opposite angles supplementary}$$

$$\therefore \tan A + \tan B + \tan C + \tan D \quad (1)$$

$$= \tan A + \tan B + \tan(180^\circ - A) + \tan(180^\circ - B)$$

$$= \cancel{\tan A} + \cancel{\tan B} - \cancel{\tan A} - \cancel{\tan B} \quad (1)$$

$$= 0$$

$$b)(i) v = 6 + Ae^{-kt}$$

$$\begin{aligned}\therefore \frac{dv}{dt} &= Ae^{-kt} \times (-k) \\ &= -kAe^{-kt} \\ &= -k(6 + Ae^{-kt} - 6) \\ &= -k(v - 6) \quad \textcircled{1} \text{ for process} \\ &= k(6 - v)\end{aligned}$$

$$\text{When } t = 0, v = 30 :$$

$$\therefore 30 = 6 + Ae^0$$

$$\therefore A = 24 \quad \textcircled{1}$$

$$(ii) v = 6 + 24e^{-kt}$$

$$\text{When } t = 1, v = 10.7$$

$$\therefore 10.7 = 6 + 24e^{-k}$$

$$\therefore \frac{4.7}{24} = e^{-k}$$

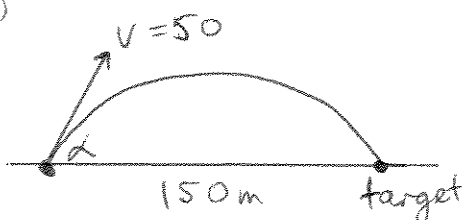
$$\therefore -k = \log\left(\frac{4.7}{24}\right)$$

$$\therefore k \doteq 1.63 \quad \textcircled{1}$$

$$(iii) v = 6 + 24e^{-3.26}$$

$$\doteq 6.9 \text{ m/s} \quad \textcircled{1}$$

c)(i)



$$x = 50 \cos \alpha t \quad \textcircled{1}$$

$$y = 50 \sin \alpha t - 5t^2 \quad \textcircled{1}$$

(ii) At target: $x = 150, y = 0$

$$\therefore 150 = 50 \cos \alpha t$$

$$\therefore \cos \alpha t = 3$$

$$\therefore t = \frac{3}{\cos \alpha} \quad \textcircled{1}$$

$$\therefore 0 = 50 \sin \alpha \cdot \frac{3}{\cos \alpha} - 5 \left(\frac{3}{\cos \alpha} \right)^2$$

$$= 150 \tan \alpha - \frac{45}{\cos^2 \alpha} \quad \textcircled{1} \text{ for process}$$

$$= 150 \tan \alpha - 45 \sec^2 \alpha$$

$$= 150 \tan \alpha - 45(1 + \tan^2 \alpha)$$

$$\therefore 45 \tan^2 \alpha - 150 \tan \alpha + 45 = 0$$

$$\therefore 3 \tan^2 \alpha - 10 \tan \alpha + 3 = 0$$

$$(3 \tan \alpha - 1)(\tan \alpha - 3) = 0 \quad \textcircled{1} \text{ for eqn}$$

$$\therefore \tan \alpha = \frac{1}{3} \text{ or } 3$$

$$\therefore \alpha = 18^\circ 26' \text{ or } 71^\circ 34' \quad \textcircled{1}$$

$\therefore$ soldier can hit target with either angle above.

7 a) (i) For $f(x) = \sec x$:

$$D: 0 \leq x < \frac{\pi}{2}$$

$$R: y \geq 1$$

$\therefore$ for $f^{-1}(x)$, $D: x \geq 1$ ①

(ii) $y = \sec x$

$$\therefore f^{-1}(x): x = \sec y \quad \text{① for process}$$

$$= \frac{1}{\cos y}$$

$$\therefore \cos y = \frac{1}{x}$$

$$\therefore y = \cos^{-1}\left(\frac{1}{x}\right)$$

$$(iii) \frac{d}{dx} \left[\cos^{-1}\left(\frac{1}{x}\right) \right] = \frac{-1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \cdot (-x^{-2})$$

$$= \frac{1}{\sqrt{1 - \frac{1}{x^2}}} \cdot \frac{1}{x^2} \quad \text{①}$$

$$= \frac{1}{x^2 \sqrt{1 - \frac{1}{x^2}}} \quad \text{①}$$

$$\left(\text{or } \frac{1}{\sqrt{x^4 - x^2}} \quad \text{or } \frac{1}{x \sqrt{x^2 - 1}} \right)$$

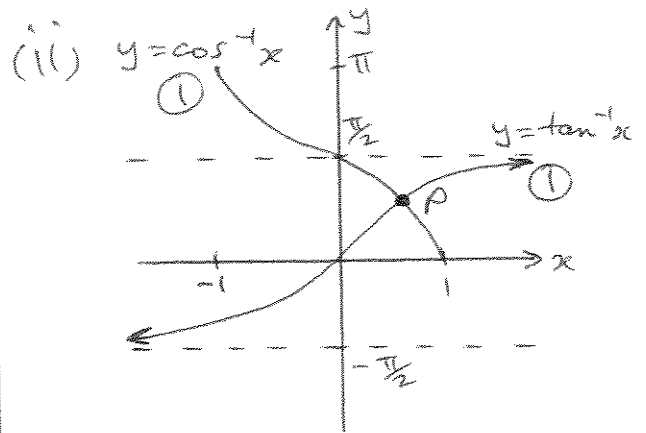
b) (i) Solve $m^2 + m - 1 = 0$ ($m = x^2$)

$$\therefore m = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times -1}}{2}$$

$$= \frac{-1 \pm \sqrt{5}}{2} \quad \text{①}$$

$$\therefore x^2 = \frac{-1 + \sqrt{5}}{2} \text{ only (as } x^2 \geq 0)$$

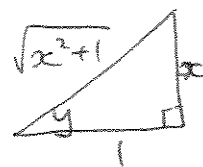
$$\therefore x \doteq \pm 0.786 \quad \text{①}$$



(-1 if x/y points missing)

(iii) At P, $\tan^{-1}x = y = \cos^{-1}x$

$$\therefore x = \tan y \quad \text{①}, \quad x = \cos y$$



$$\therefore \cos y = \frac{1}{\sqrt{x^2 + 1}}, \quad \tan y = \frac{\sqrt{1 - x^2}}{x} \quad \text{①}$$

Now, since $\tan y = \cos y (= x)$

$$\text{then either: } \frac{1}{\sqrt{x^2 + 1}} = \frac{\sqrt{1 - x^2}}{x}$$

$$\text{or } \frac{1}{\sqrt{x^2 + 1}} = x \quad \text{①}$$

$$\text{or } \frac{\sqrt{1 - x^2}}{x} = x$$

which all give $x^4 + x^2 - 1 = 0$

Coordinates of P are

$$(0.786, 0.666)$$

①